Multistage Hybrid Denoising Network for Satellite Hyperspectral Images

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***Abstract*— The hyperspectral imaging instrument makes a trade-off by sacrificing spatial resolution to achieve high spectral resolution. This compromise leads to a low signal-to-noise ratio, and hyperspectral images (HSIs) are often heavily contaminated with mixed noise, which is an inherent challenge. Previous research has achieved satisfactory results for natural image denoising; hyperspectral denoising has remained a formidable task. In this article, we introduce an innovative method called the multistage hybrid-denoising network for satellite hyperspectral images (SUC-MSDN). SUC-MSDN initially decomposes the noisy HSI into multiple scales and constructs a multistage denoising network by analyzing the spatial spectrum texture distribution characteristics of noise signals. Instead of simply stacking the output results from each scale, SUC-MSDN uses the denoising results from the low-scale network as prior knowledge for the high-scale denoising network to more accurately remove the final noise components. Extensive experimental datasets are used to validate the performance of SUC-MSDN. Experimental results show that SUC-MSDN outperforms benchmark methods and significantly enhances the accuracy of land cover mapping.**

***Index Terms*— Deep learning, hyperspectral image (HSI), multistage denoising, multistage hybrid denoising network for satellite hyperspectral images (SUC-MSDN).**

I. Introduction

HYPERSPECTRAL satellite sensors capture continuous surface reflectance across the spectrum from 400 to 2500 nm [1]. The increasing availability of hyperspectral image (HSI), exemplified by initiatives such as the launch of the EO-1 Hyperion hyperspectral imaging spectrometer by the United States in 2000, the GF-5 hyperspectral imaging spectrometer by China in 2018, and the ZY-1 02 D hyperspectral imaging spectrometer by China in 2019, has led to widespread applications in environmental monitoring and resource exploration [2]. However, hyperspectral sensors, due to their narrow spectral bands, capture limited surface reflected energy, making them highly susceptible to environmental factors such as weather conditions [3], [4], [5]. As a result, HSI are often contaminated by mixed noise signals, including Gaussian noise, impulse noise, stripes, and deadlines [6]. Because noise disrupts the continuity of surface reflectance signals, the application performance of HSI has been severely degraded. Therefore, the removal of mixed noise is imperative to enhance the practicality and reliability of HSI [7].

To address the challenge of removing mixed noise from HSI, an enormous amount of research has been developed [8]. These methods can be roughly categorized into 1) filter-based methods; 2) matrix decomposition-based methods; 3) deep learning-based methods; and 4) hybrid methods [9], [10]. Filter-based methods typically commence by analyzing the noise distribution characteristics and then removing noise signals through thresholding techniques [11]. These methods offer significant computational efficiency. However, some valuable signals are removed, which results in the image being overly smooth and having ringing artifacts [3]. In recent years, matrix decomposition theory has been widely applied in image processing, especially image denoising. These techniques assume that clean components in HSI lie within a linear subspace with a dimension significantly smaller than the number of spectral bands. While matrix decomposition techniques have significantly improved HSI denoising performance, these methods often involve numerous parameters, exhibit limited robustness, and can be computationally inefficient [3]. Recently, motivated by the success of deep learning in nonlinear modeling, deep nonlinear HSI denoising networks have been extensively researched [12], [13]. These methods have become a research hotspot in HSI denoising. Additionally, some researchers have focused on combining matrix decomposition methods with deep learning approaches to propose hybrid methods. These methods aim to enhance the denoising performance of model while also improving the interpretability of deep learning.

It is worth noting that compared to other methods, supervised deep learning methods, especially multiscale decomposition methods, have gained a significant advantage for hyperspectral denoising. Nevertheless, there are persistent issues that require attention. For example, mixed denoising methods extract multiscale features from HSI by adjusting

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dilation rates in convolutions or by employing convolutional down-sampling and up-sampling features, similar to U-Net. Instead of exploring how multiscale features can effectively collaborate for denoising, these methods often resort to simply concatenating multiscale features as inputs for subsequent layers. This simplistic method underutilizes the wealth of multiscale information and substantially increases computational requirements.

To address the challenges of HSI denoising, we propose a novel multistage hybrid-denoising network for satellite hyperspectral images (SUC-MSDN). This network employs a multistage training strategy to enhance denoising performance. Specifically, SUC-MSDN begins by decomposing the noisy HSI into multiple scales. In the initial stage, the lower-scale image is trained, and the acquired knowledge (the denoised features obtained on low-scale image) is used as prior information to improve noise feature extraction in subsequent stages. At each stage, SUC-MSDN explores the spatial–spectral characteristics and texture characteristics of noise distribution, and a novel noise attention module (NAM) is constructed for optimizing noise features. The main contributions of this article are as follows.

1) We propose the SUC-MSDN. Instead of feature stacking, SUC-MSDN establishes connections between outputs and input data at different scales by exploring the mutually reinforcing characteristics of features, which improved denoising performance and computational efficiency.
2) We have constructed the spatial–spectral-texture feature extraction module (SSTE) to effectively explore the spatial–spectral and texture distribution characteristics of noise signals. The design of the NAM significantly enhances the noise extraction capabilities. Additionally, the proposed multiscale connectivity module (MSCB) enhances the interaction between features at different scales. Furthermore, the design of the feature reconstruction module (FRM) connects noise signals across different scales, improving the final noise extraction accuracy.
3) SUC-MSDN effectively removes noise signals in both simulated and real datasets, demonstrating competitive denoising performance and superior computational efficiency compared to nine advanced methods.

The structure of the article is as follows: Section II describes the related work. Section III provides an overview of the framework and network details. Section III presents experimental results on both simulated and real datasets. Section IV provides the conclusion.

II. Related Work

Traditional filtering methods include Gaussian filtering [14], median filtering [15], adaptive Wiener filtering [16], and others [17]. Nevertheless, these methods often struggle with the delicate balance between preserving image fidelity and achieving effective denoising. Considering the frequency distribution characteristics of noise signals, techniques such as Fourier transformation and wavelet decomposition theory have been used for image denoising [18]. Jain and Tyagi [19] proposed an innovative edge-preserving image denoising method based on wavelet transformation. Sami et al. [20] used a robust median estimator to measure the noise-to-signal ratio of noisy images, proposing a novel image denoising method grounded in discrete wavelet transformation. Bai and Feng [21] have proposed a novel fractional-order anisotropic diffusion equation for noise removal, implementing a numerical algorithm using discrete Fourier transformation and presenting an iterative scheme in the frequency domain. To exploit the sparse features of noise, Sarjanoja et al. [22] proposed a novel 2-D image denoising strategy based on transform-domain enhanced sparse representation by grouping 2-D image patches into 3-D data arrays. To explore the spatial-spectral sparsity of 3-D images, Maggioni et al. [23] employed voxel cubes arranged in a 4-D group. The 4-D transformation applied to this group simultaneously leveraged local correlations between voxels within each cube and nonlocal correlations between corresponding voxels in different cubes, rendering the group spectrum highly sparse and facilitating effective noise separation through coefficient shrinkage.

In matrix decomposition-based methods, low-rank decomposition [24], sparse representation [25], and tensor decomposition [26] have demonstrated their effectiveness. Recognizing that noise levels can vary across different spectral bands in HSI, He et al. [27] proposed the noise-adjusted iterative low-rank matrix approximation (NAILRMA) method for HSI denoising. This method establishes a piecewise low-rank matrix approximation (LRMA) model and an iterative regularization framework is used to solve the objective equation. Wang et al. [28] proposed a low-rank matrix recovery algorithm for image denoising that combines nuclear norm, total variation (TV) norm, and l_1-norm to explore the low-rank properties of images, and uses an iterative alternating direction and fast gradient projection method to solve the nonconvex optimization problem. Understanding that noise in HSI can be sparsely represented by a finite number of atoms, Zeng et al. [29] introduced the dictionary pair learning on Grassmann manifold (DPLG) algorithm. DPLG learns an initial dictionary pair on the Grassmann manifold using subspace-partitioning techniques and then encodes each image block using the selected sub dictionary pairs to obtain a sparse representation. The nonzero elements of the sparse representation are further smoothed by the graph Laplacian operator to remove noise. Lu et al. [30] proposed the spectral–spatial adaptive sparse representation (SSASR) method for HSI denoising. SSASR divides the HSI into nonoverlapping spectral band subsets based on interband correlation. Within each subset, it identifies shape-adaptive local regions composed of spatially similar pixels and exploits highly correlated and similar spectral–spatial information in each group through joint sparse coding to generate a better noise-free estimation. Considering the 3-D structural characteristics of HSI, tensor decomposition has proved highly effective in describing the spatial–spectral distribution of noise. Xue et al. [31] proposed a nonlocal low-rank regularized CANDECOMP/PARAFAC [canonical polyadic (CP)] tensor decomposition model (NLR-CPTD), merging spectral global correlation, and spatial local self-similarity prior. Zhang et al. [32] combined nonlocal low-rank tensor decomposition (NLRTD) with TV regularization, presenting TV-NLRTD. Wu et al. [33] grouped

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similar patches into 3-D stacks and transformed the stacks into a finite sum of rank-1 products, proposing an efficient CP decomposition algorithm.

Deep learning has achieved significant success in image denoising. These methods achieve denoising by learning the nonlinear mapping relationship between clean and noisy HIS, primarily include convolutional neural network (CNN), generative adversarial networks (GANs), transformer, and diffusion models. Yuan et al. [34] proposed a spatial–spectral deep CNN (HSID-CNN) for HSI denoising. Maffei et al. [35] introduced the hyperspectral single denoising CNN (HSISDeCNN). Zhang et al. [36] proposed an end-to-end HSI denoising model based on adversarial learning. Specifically, to capture subtle noise distributions from both spatial and spectral dimensions, a residual spatial–spectral module (RSSM) was designed and embedded into a U-Net-like structure as the generator to produce clear images. To distinguish between real and generated images, a discriminator based on a multiscale feature fusion module (MFFM) was developed to further enhance the quality of the denoising results [36]. Li et al. [37] presented a spectral–enhanced rectangular transformer network that explores the nonlocal spatial similarity and global spectral low-rank characteristics of HSI. Miao et al. [38] proposed a self-supervised diffusion model for HSI restoration, known as the denoising diffusion spatiotemporal spectral model (DDS2M). This model infers the parameters of the proposed variational spatiotemporal spectral module (VS2M) during the reverse diffusion process without requiring additional training data [38].

To leverage the advantages of various theoretical approaches, some hybrid methods combine matrix decomposition with deep learning for image denoising. Sun et al. [39] proposed an HSI denoising model that integrates sparse low-rank representation with CNNs. Miao et al. [40] introduced an unsupervised deep image prior (DIP) framework based on classic HSI spatiotemporal spectral decomposition. Xiong et al. [41] proposed an end-to-end model-assisted nonlocal neural network (MAC-Net), which simultaneously considers spectral low-rank models and spatial deep priors to reduce HSI noise. Additionally, Xiong et al. [42] proposed a novel model-guided HSI denoising interpretable network, which considers spatial redundancy, spectral low-rank property, and spectral–spatial correlations.

III. Methodology

A. Ground Truth Generation Strategy

In our research, we treat unmanned aerial vehicle (UAV) datasets as clean images because they are less affected by atmospheric interference [43]. These clean images are divided into 128 × 128 × 25 pixels patches. In the patch extraction process, we use a spectral step size of 5 and a spatial step size of 64, significantly increasing the number of samples. Because UAV data and satellite HSI may differ in imaging and spectral resolution, we randomly rearrange the spectral order of these patches. This step helps prevent limited network generalization performance due to spectral distribution differences. Subsequently, we simulate noise signals of different types and intensities, and add these noises to clean patches to generate noisy samples. In the end, we use 16 000 clean-noisy sample pairs for network training, 1860 sample pairs for network validation, and 100 sample pairs for network testing. We generate five different noise mixing scenarios, which are detailed as follows:

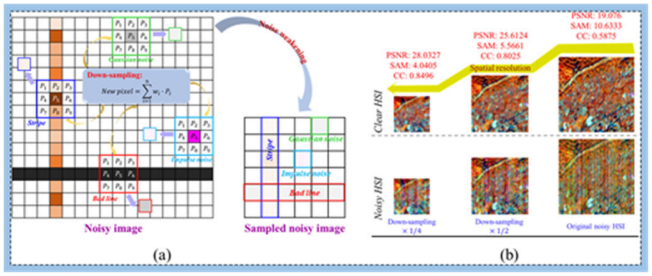


Fig. 1. Effect of downscaling on noisy HSI. (a) Process of image downscaling. (b) Quality of noisy HSI at different scales.

1) *Case 1 (Gaussian Noise):* Gaussian noise is simulated and added to all bands of the clean data. It is important to note that the noise intensity in each band of the simulated noisy image varies. The intensity of Gaussian noise is set to [5–30 dB].
2) *Case 2 (Gaussian Noise + Impulse Noise):* Gaussian noise and impulse noise are simulated and added to all bands of the clean data. The intensity of Gaussian noise is set to [5–30 dB], while the intensity of impulse noise is set to [0.05–0.15].
3) *Case 3 (Gaussian Noise + Impulse Noise + Stripes):* Gaussian noise and impulse noise are added to each band of the clean image, with intensities set the same as in Case 2. Additionally, [5%–30%] of the bands are randomly selected, and [5%–30%] columns of pixels are randomly selected to add stripes in these bands.
4) *Case 4 (Gaussian Noise + Impulse Noise + Deadlines):* Gaussian noise and impulse noise settings are the same as in Case 2. Furthermore, [5%–30%] of the bands are selected, and [5%–30%] columns of pixels are randomly selected to add deadlines in these bands.
5) *Case 5 (Gaussian Noise + Impulse Noise + Stripes + Deadlines):* Gaussian noise and impulse noise settings are the same as in Case 2, stripes settings are the same as in Case 3, and deadlines settings are the same as in Case 4.

B. Multistage Hybrid Denoising Network for Satellite Hyperspectral Images

The observed noisy HSI $\mathbf{Y} \in \mathbb{R}^{m\times n\times k}$ can be obtained by adding a clean HSI $\mathbf{X} \in \mathbb{R}^{m\times n\times k}$ to a noise image $\mathbf{N} \in \mathbb{R}^{m\times n\times k}$, where m and n represent the height and width of the image, respectively, and k represents the number of channels. Therefore, the HSI degradation model can be represented as

$$\mathbf{Y} = \mathbf{X} + \mathbf{N}. \tag{1}$$

Based on the above model, deep learning methods aim to extract $\mathbf{N}$ from $\mathbf{Y}$ using a nonlinear mapping and obtain the final $\mathbf{X}$ by subtracting $\mathbf{N}$ from $\mathbf{Y}$. To design a deep hyperspectral mixed denoising model, extensive research has been proposed, and the extraction of multiscale features is considered effective in reconstructing $\mathbf{X}$ [44]. The reason is

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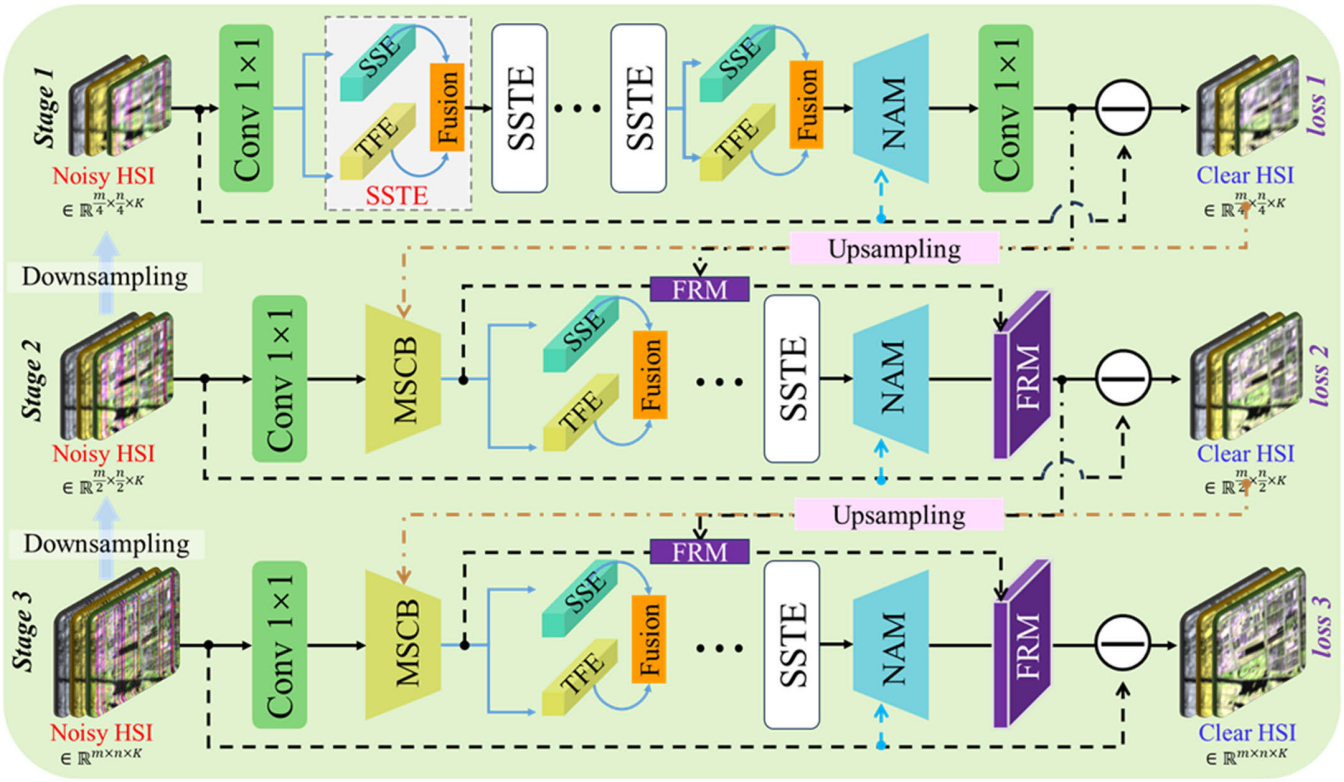


Fig. 2. Multistage hybrid denoising network for satellite hyperspectral images.

Algorithm 1 Training SUC-MSDNN

Input data: Noise image $\mathbf{Y}$
Output results: Network parameters

Data processing: Perform multiscale decomposition on $\mathbf{Y}$ to obtain $\mathbf{Y}_1, \mathbf{Y}_2, \mathbf{Y}_3$.
for i = 1:epoch
Stage-1: Calculate $\mathbf{X}_1$ (Equation (2)–(3))
a. Extract and fuse spatial spectral texture features from $\mathbf{Y}_1$ to obtain $\mathbf{N}_1^{in}$ (Equation (2)).
b. Using NAM to optimize $\mathbf{N}_1^{in}$ and subtracting $\mathbf{N}_1$ from $\mathbf{Y}_1$ to obtain $\mathbf{X}_1$ (Equation (3)).
Stage-2: Calculate $\mathbf{X}_2$ (Equation (4)–(7))
c. $(\mathbf{X}_1)\uparrow$ assists $\mathbf{Y}_2$ for noise separation in MSCB to obtain $\mathbf{N}_j^{in}$ (Equation (4))
d. Extract and fuse the features of $\mathbf{N}_j^{in}$, and use NAM for feature optimization to obtain $\mathbf{N}_j^{op}$ (Equation (5))
e. Fuse multiscale noise features to obtain $\mathbf{N}_2$ (Equation (6))
f. Subtract $\mathbf{N}_2$ from $\mathbf{Y}_2$ to obtain $\mathbf{X}_2$ (Equation (7))
Stage-3: Calculate $\mathbf{X}_3$ (Perform the same operation as Stage 2 after replacing the input data)
Loss calculation: Optimize network parameters
end

that $\mathbf{Y}$ merges the pixel values of neighboring pixels to obtain new pixel values during the downscaling process. When the pixel values of neighboring pixels are not contaminated by noise or are lightly contaminated, the noise signal contained in the new pixel values will be weakened [see Fig. 1(a)]. Therefore, as shown in Fig. 1(b), downscaled images typically have greater similarity to the clear image and are more conducive to separating noise signals. Unfortunately, many existing multiscale deep denoising networks simply stack multiscale image features without delving into the underlying mechanisms of their interactions [45]. These methods reduce network interpretability, potentially leading to suboptimal performance in multiscale tasks. Second, working with high-dimensional features imposes a substantial computational burden on the network, which can be particularly problematic when dealing with high-dimensional HSI data.

Fig. 2 illustrates the network framework of SUC-MSDN. The SUC-MSDN model first applies a multiscale decomposition to $\mathbf{Y}$ using a downsampling factor of 2, obtaining the multiscale dataset $\mathbf{Y} = [\mathbf{Y_1}, \mathbf{Y_2}, \mathbf{Y_3}]$. In the first stage, $\mathbf{Y_1}$ is used as the input data. After a 1×1 convolution, spatial–spectral features and texture features are extracted using spatial–spectral feature extraction module (SSE) and texture feature extraction module (TFE), respectively, and these features are then fused. After five stages of spatial–spectral-texture feature extraction and fusion, the initial noise $\mathbf{N}_1^{\text{in}}$ is obtained. Subsequently, NAM is constructed to optimize the noise, yielding $\mathbf{N_1}$. Finally, the clear small-scale image $\mathbf{X_1}$ is obtained by subtracting $\mathbf{N_1}$, after applying a 1×1 convolution, from $\mathbf{Y_1}$. The specific formula is as follows:

$$\begin{aligned} &\mathbf{N}_1^{\text{in}} = (P_i, \ldots, (P_1(\mathscr{b}_1(\mathbf{Y_1})))), \quad i = [1, 2, 3, 4, 5] \\ &\text{s.t. } P = G_f(V_s(I), V_t(I)) \end{aligned} \tag{2}$$

$$\begin{aligned} &\mathbf{X_1} = \mathbf{Y_1} - \mathbf{N_1} \\ &\text{s.t. } \mathbf{N_1} = \mathscr{b}_1\left(A_\theta\left(\mathbf{N}_1^{\text{in}}\right)\right) \end{aligned} \tag{3}$$

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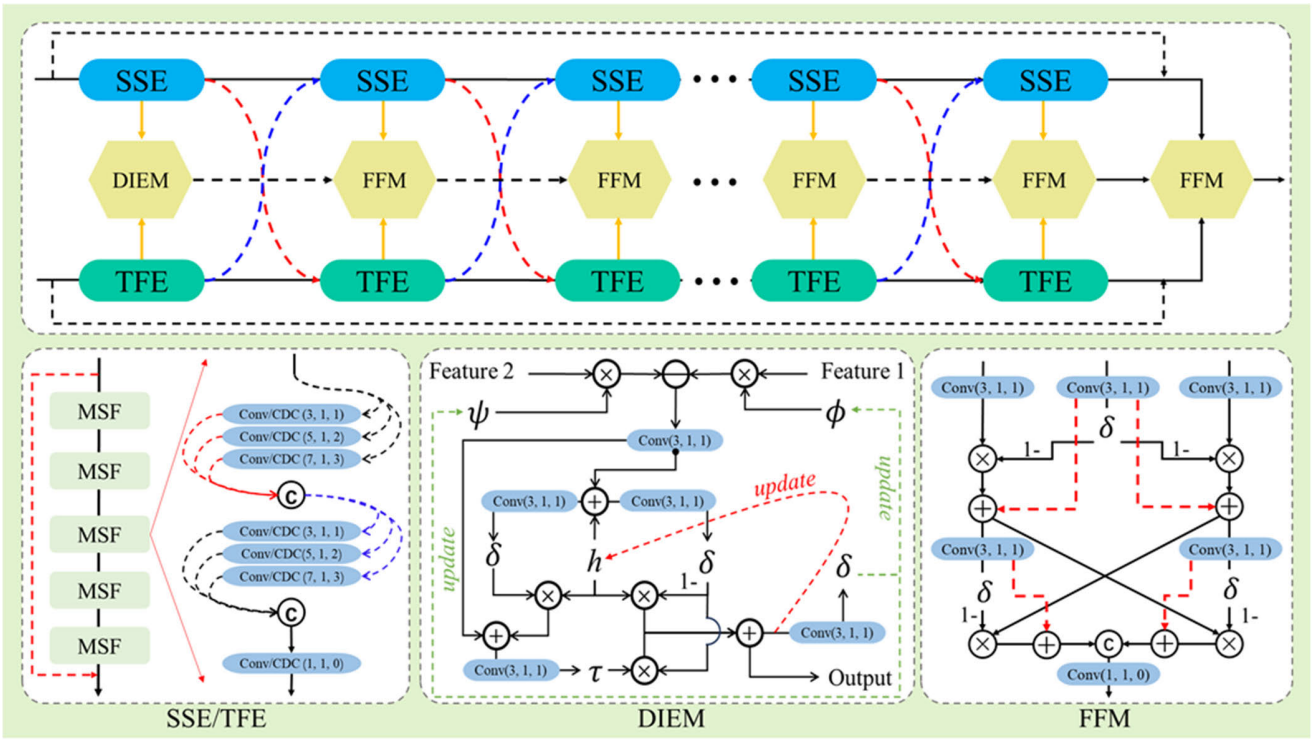


Fig. 3. Spatial–spectral-texture feature extraction module.

where $\mathscr{C}_1$ represents the 1×1 convolution, A_θ denotes the NAM operation, V_s represents the SSE operation, V_t represents the TFE operation, and G_f refers to the feature fusion operation.

In the second stage, we first construct the MSCB. $\mathbf{Y_2}$, after a 1×1 convolution, is used as the input data, and the upsampled $\mathbf{X_1}$ is used as auxiliary data to initially separate the medium-scale noise feature $\mathbf{N}_2^{\text{in}}$. To further optimize the noise, $\mathbf{N}_2^{\text{in}}$ undergoes five stages of SSE and TFE operations and is then input into the NAM to obtain the optimized noise feature $\mathbf{N}_2^{\text{op}}$. After that, $\mathbf{N_1}$ is upsampled and fused with $\mathbf{N}_2^{\text{in}}$ and $\mathbf{N}_2^{\text{op}}$ in the FRM module to obtain $\mathbf{N_2}$. Finally, subtracting $\mathbf{N_2}$ from $\mathbf{Y_2}$ yields the medium-scale clear image $\mathbf{X_2}$. The third stage follows the same process as the second stage. Specifically, the upsampled $\mathbf{X_2}$ is used as auxiliary data to initialize the separation of noise features from $\mathbf{Y_3}$, and $\mathbf{N_2}$ is upsampled and fused with $\mathbf{N}_3^{\text{in}}$ and $\mathbf{N}_3^{\text{op}}$ in the FRM module to obtain $\mathbf{N_3}$. Finally, subtracting $\mathbf{N_3}$ from $\mathbf{Y_3}$ produces the large-scale clear image $\mathbf{X_3}$. The specific formula can be expressed as follows:

$$\mathbf{N}_j^{\text{in}} = MS_\theta\left(\mathscr{C}_1\left(\mathbf{Y}_j\right), \left(\mathbf{X}_{j-1}\right)\uparrow\right), \quad j = [2, 3] \tag{4}$$

$$\mathbf{N}_j^{\text{op}} = A_\theta\left(P_i, \ldots, \left(P_1\left(\mathbf{N}_j^{\text{in}}\right)\right)\right), \quad i = [1, 2, 3, 4, 5] \tag{5}$$

$$\mathbf{N}_j = W_f\left(\mathbf{N}_j^{\text{in}}, \mathbf{N}_j^{\text{op}}, \left(\mathbf{N}_{j-1}\right)\uparrow\right) \tag{6}$$

$$\mathbf{X}_j = \mathbf{Y}_j - \mathbf{N}_j \tag{7}$$

where MS_θ represents the MSCB operation, and j denotes the jth scale. $\uparrow$ indicates the upsampling operation.

C. Network Architecture

1) Spatial–Spectral-Texture Feature Extraction Module: As shown in Fig. 3, the SSTE module consists primarily of four components: SSE, TFE, difference information extraction module (DIEM), and feature fusion module (FFM).

HSI contains rich spatial–spectral information, which not only describes information such as the distance, direction, and shape of targets, but also manifests the electromagnetic spectral reflection characteristics of targets. In addition, HSI also includes abundant texture information, describing the grayscale or color distribution of pixels in local regions of the image. However, spatial–spectral information of objects is disrupted by noise contamination, causing changes in edge shapes and nonsmooth spectral curves, while texture information is degraded by abrupt changes in pixel values within regions. Therefore, simultaneously exploring the spatial–spectral distribution characteristics and texture distribution characteristics of noise signals in HSI would benefit noise signal separation and improve the denoising accuracy of algorithms. Accordingly, in our experiments, we employed convolution operations to explore the spatial–spectral distribution patterns of noise, and central difference convolution operations to explore the texture distribution patterns of noise. Fig. 3 illustrates the construction of SSTE. The SSTE consists of two branches of residual networks, constructed by convolution operations and central difference convolution operations, respectively. To reduce redundancy between features, after the extraction of spatial–spectral and texture features in the first layer, the SSTE constructs a DIEM to extract differential information between features. This differential information is then used to guide subsequent convolution and central difference convolution operations. Additionally, we constructed an FFM to fuse spatial–spectral and texture features. Finally, the spatial–spectral features, texture features, and fused features

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are integrated again to obtain the final output information. Specific formulas can be defined as follows:

$$S_f^1 = S_1(\mathbf{NI}), \quad T_f^1 = T_1(\mathbf{NI}) \tag{8}$$

$$F_d = D\left(S_f^1, T_f^1\right) \tag{9}$$

$$F_f^1 = \mathcal{F}\left(T_f^1 + F_d, S_f^1 + F_d, F_d\right) \tag{10}$$

$$\cdots \tag{11}$$

$$F_f^n = \mathcal{F}\left(S_f^{n-1} + F_f^{n-2}, T_f^{n-1} + F_f^{n-2}, F_f^{n-2}\right) \tag{12}$$

where **NI** denotes the noisy image, and S and T represent the convolution and central difference convolution operations, respectively. S_f and T_f are the spectral features and texture features, respectively. D represents DIEM, F represents FFM, F_d denotes differential information, and F_f signifies fused information. The notation n indicates the nth layer of the convolution operation.

For SSE and TFE, we constructed a residual network consisting of five multiscale feature extraction modules. In each multiscale feature extraction module, we perform 3×3, 5×5, and 7×7 convolutions or central difference convolutions on the input data. After stacking these multiscale features along the channel dimension, we perform multiscale operations again. Finally, these features are fed into a 1×1 convolution or central difference convolution layer to obtain the final features. Each multiscale feature extraction module is computed as follows:

$$M_i = \begin{cases} \mathcal{T}_1 = c(\mathscr{b}_3(I), \mathscr{b}_5(I), \mathscr{b}_7(I)) \\ \mathcal{T}_2 = \mathscr{b}_3(c(\mathscr{b}_3(\mathcal{T}_1), \mathscr{b}_5(\mathcal{T}_1), \mathscr{b}_7(\mathcal{T}_1))) \end{cases} \tag{13}$$

where I denote the input data, $\mathscr{b} = [S, T]$, and $\mathscr{b}_3$, $\mathscr{b}_5$, and $\mathscr{b}_7$ represent the 3×3, 5×5, and 7×7 convolutions or central difference convolutions, respectively. The symbol c denotes the feature stacking operation along the channel dimension. T denotes the output features, and M_i represents the ith multiscale feature extraction module.

In DIEM, we constructed a gating mechanism to precisely extract the differential information between features. DIEM first constructs two initial unit matrices, ψ and ϕ, and multiplies them with the input features. The differential information is then obtained by subtracting the features. This differential information is input into the gating network to obtain the output features. Notably, these output features are iteratively fed back to the state h for further computation. After the iterative computation on the state h is completed, DIEM uses a sigmoid activation function to analyze the distribution of the output features and updates ψ and ϕ, multiplying them with the input data to reenter the iterative computation. It is worth mentioning that DIEM can continuously update its internal state, improving the accuracy of the output features. Additionally, it optimizes the input data using the output features to further enhance the accuracy of differential information extraction. DIEM can be represented as follows:

$$F_d^{j-1} = \psi^{j-1} \times F_1 - \phi^{j-1} \times F_2 \tag{14}$$

$$z_i = \sigma\left(w_z \times \left[h_{i-1}, F_d^{j-1}\right]\right) \tag{15}$$

$$r_i = \sigma\left(w_r \times \left[h_{i-1}, F_d^{j-1}\right]\right) \tag{16}$$

$$\tilde{h}_i = \tanh\left(w \times \left[r_i \odot h_{i-1}, F_d^{j-1}\right]\right) \tag{17}$$

$$h_i = (1 - z_i) \odot h_{i-1} + z_i \odot \tilde{h}_i \tag{18}$$

$$\psi^j = \sigma(h_i), \quad \phi^j = \sigma(h_i) \tag{19}$$

where j denotes the jth update of ψ and ϕ, F_d represent the differential information, i denotes the ith update of the state h, h is the hidden state, $\tilde{h}$ is the candidate hidden state, r is the reset gate, z is the update gate, σ is the sigmoid activation function, and tanh is the tanh activation function.

To further de-redundant and fuse the features from different branches, we also constructed FFM. FFM first applies a sigmoid activation to the fused features from the previous layer and subtracts them from the unit matrix. The result is then multiplied by the SSE and TFE features to obtain the unique features of SSE and TFE. After this, the fused features are added to the unique features of SSE and TFE separately to obtain the new features S_f^s and T_f^s. Then, S_f^s and T_f^s are again activated by sigmoid and subtracted from the unit matrix, and multiplied with S_f^s and T_f^s, respectively. Finally, these features are added to the original S_f^s and T_f^s. S_f^s and T_f^s are stacked on the channel dimension, and passed through a 1×1 convolution operation to obtain the final fused features. This process is represented as follows:

$$S_{sf}^i = \left(1 - \sigma\left(\mathscr{b}_3\left(F^i\right)\right)\right) \times \mathscr{b}_3\left(S_f^i\right) + \mathscr{b}_3\left(F^i\right) \tag{20}$$

$$T_{sf}^i = \left(1 - \sigma\left(\mathscr{b}_3\left(F^i\right)\right)\right) \times \mathscr{b}_3\left(T_f^i\right) + \mathscr{b}_3\left(F^i\right) \tag{21}$$

$$ST^i = \left(1 - \sigma\left(\mathscr{b}_3\left(S_{sf}^i\right)\right)\right) \times \mathscr{b}_3\left(T_{sf}^i\right) + \mathscr{b}_3\left(S_{sf}^i\right) \tag{22}$$

$$TS^i = \left(1 - \sigma\left(\mathscr{b}_3\left(T_{sf}^i\right)\right)\right) \times \mathscr{b}_3\left(S_{sf}^i\right) + \mathscr{b}_3\left(T_{sf}^i\right) \tag{23}$$

$$F_f^i = \mathscr{b}_1\left(c\left(ST^i, TS^i\right)\right) \tag{24}$$

where F denotes the fused features, i represents the ith layer, and S_{sf}^i, T_{sf}^i, ST^i, and TS^i are the output features. F_f represents the final output features. Overall, in the first stage, to reduce the redundancy between spatial–spectral features and texture features relative to the differential information, FFM initially utilizes the sigmoid function to explore the distribution characteristics of the differential information. It then uses these distribution characteristics to separately optimize the spatial–spectral features and texture features, enhancing the heterogeneous information that is distinct from the differential information. In the second stage, aiming to reduce the informational redundancy between spatial–spectral features and texture features themselves, the sigmoid function is applied to separately explore their distribution characteristics. Through cross-multiplication, this process enhances the heterogeneous information between the two feature types.

2) Noise Attention Module: During the noise extraction process, the complex mixture of noise signals and target signals often results in the initially extracted noise features containing valuable target information, leading to the loss of some critical details in the denoised image. To optimize the extracted noise features, we designed NAM (see Fig. 4). NAM first applies a 1×1 convolution operation to the extracted noise features and subtracts the noise features from the noisy image to obtain a clear image. To analyze the information distribution of the clear features, the clear image undergoes a 1×1 convolution operation and is then activated by a

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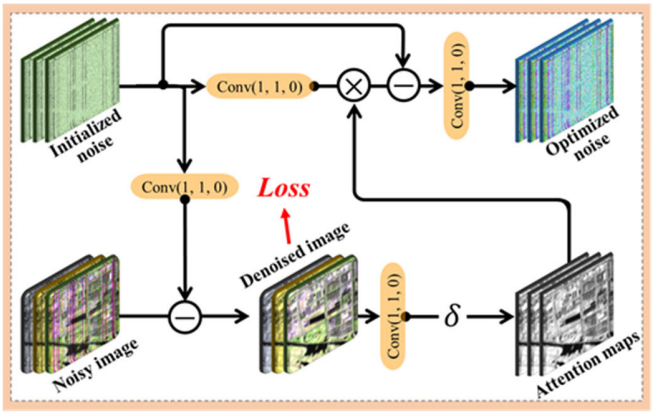


Fig. 4. Noise attention module.

sigmoid function. The activated matrix is multiplied by the original noise features to extract the clear features mixed in the noise features. Finally, the initially extracted noise features are optimized by subtracting the mixed clear features. It is worth noting that we calculate the loss on the clear image produced in the process to ensure that the information distribution of the clear features obtained through the sigmoid activation is reasonable. NAM can be specifically represented as follows:

$$\mathcal{A}_c = \sigma\left(\mathscr{b}_1\left(N_{\text{image}} - N_{\text{in}}\right)\right) \tag{25}$$

$$N_{\text{op}} = \mathscr{b}_1(N_{\text{in}} - \mathscr{b}_1(N_{\text{in}}) \times \mathcal{A}_c) \tag{26}$$

where N_{image} denotes the noisy image, N_{in} is the initial noise features, $\mathcal{A}_c$ is the attention map, and N_{op} is the final optimized noise features.

3) Multiscale Connectivity Module: In previous methods [34], [46], multiscale features in the network were simply stacked, resulting in computational burdens and a lack of synergy among these features for precise image denoising, which also reduced interpretability. In SUC-MSDN, the output from the previous stage network represents a clear image on a smaller scale, while the input for the subsequent stage network is a noisy image, similar to data at different time points. This transformation allows us to treat noise separation as a differential feature extraction problem. However, there are differences not only in noise distribution but also in spatial information between the previously up-sampled small-scale clear image and the subsequent input data at another stage (the reason is that small-scale denoised outputs still have some errors). A straightforward feature subtraction may have resulted in error accumulation across different stages. To solve this problem, we constructed MSCB to achieve information transfer across different scales, using small-scale clear features as prior information to effectively separate large-scale noise signals (see Fig. 5). MSCB primarily includes two branch convolutional networks, and an information extraction module (IEM) is embedded between the same convolution layers of the two branch networks to extract noise features. This can be specifically represented as follows:

$$F_c^u = LC1\left(\mathscr{b}_3\left(F_c^{n-1}\right), u\right) \tag{27}$$

$$F_N^n = LC2\left(\mathscr{b}_3\left(F_{ni}^n\right), F_c^u\right) \tag{28}$$

where F_c^u denotes the output features, F_N^n is the separated noise signals, n represents the nth scale, $LC1$ and $LC2$ denote the Conv-LSTM cells, and u is the initial hidden state. Each Conv-LSTM cell can be represented as follows:

$$f_t = \delta(w_f \mathcal{Z}_t + w_{uf} \otimes u_{t-1} + w_{cf} \odot C_{t-1} + b_f) \tag{29}$$

$$i_t = \delta(w_i \mathcal{Z}_t + w_{ui} \otimes u_{t-1} + w_{ci} \odot C_{t-1} + b_i) \tag{30}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tanh(w_c \otimes \mathcal{Z}_t + w_{uc} \otimes u_{t-1} + b_c) \tag{31}$$

$$O_t = \delta(w_o \otimes \mathcal{Z}_t + w_{uo} \otimes u_{t-1} + w_{co} \odot C_t + b_o) \tag{32}$$

$$u_t = O_t \odot \tanh(C_t) \tag{33}$$

where $\mathcal{Z}$ represents the input data at current time step, w represents weights, b represents biases, u is the hidden output, C is the cell output, f is the forget gate, O represents the output gate, δ represents the sigmoid function, tanh represents the tanh function, and t represents the tth time step.

4) Feature Reconstruction Module: As shown in Fig. 6, to fuse noise features and generate more precise noise features, we constructed FRM. FRM first stacks noise features from different sources along the channel dimension. After this, ResNet network with eight convolutional layers is constructed to preliminarily fuse the noise features

$$N_f = \mathcal{R}(c(N_1, N_2)) \tag{34}$$

where N_1 and N_2 represent noise features from different sources, $\mathcal{R}$ denotes the ResNet network, and N_f is the preliminarily fused noise features. To achieve deep feature fusion and accurately reconstruct noise features, we perform separation operations on the preliminarily fused noise features and apply a self-attention mechanism to fuse the two components. This approach effectively enhances computational efficiency, reduces redundant information, and improves the accuracy of noise feature reconstruction. The specific formula can be represented as follows:

$$\left[N_{f1}, N_{f2}\right] = \text{split}(N_f) \tag{35}$$

$$Q = \mathscr{b}_1\left(N_{f1}\right), \quad K = \mathscr{b}_3\left(\mathscr{b}_1\left(N_{f1}\right)\right)$$

$$V = \mathscr{b}_3\left(\mathscr{b}_1\left(N_{f2}\right)\right), \quad A = \mathscr{b}_1\left(N_{f2}\right) \tag{36}$$

$$N_{fn} = A + Q \times \varphi(K \times V) \tag{37}$$

where split represents the channel separation operation, N_{f1} and N_{f2} are the output features, and Q, K, V, and A denote the convolution operation features. N_{fn} is the fused noise feature.

D. Loss Function

To ensure that SUC-MSDN achieves optimal denoising results, both spatial–spectral loss ($\mathcal{L}_{\text{sae}}$) and frequency loss ($\mathcal{L}_f$) are designed for network training, and the final loss function can be defined as

$$\mathcal{L}_{\text{loss}} = \mathcal{L}_{\text{sae}} + \mathcal{L}_f \tag{38}$$

where $\mathcal{L}_{\text{loss}}$ represents the final loss value. For $\mathcal{L}_{\text{sae}}$, the mean squared error (MSE) formula is used to calculate the error

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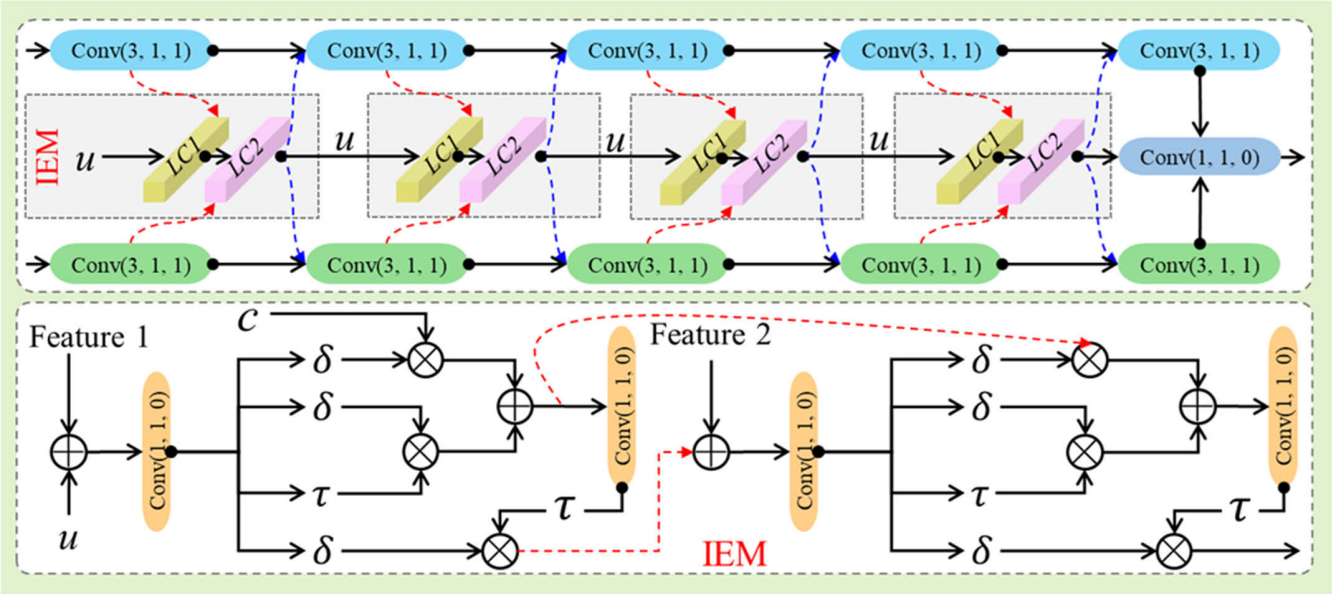


Fig. 5. Multiscale connectivity module.

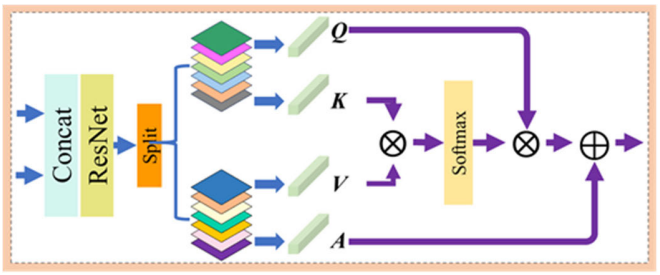


Fig. 6. Feature reconstruction module.

between the denoised result and the ground truth. This can be defined as

$$\mathcal{L}_{\text{sae}} = \sum_{i=1}^{n} ((\mathbf{Y}, \Theta) - \mathbf{Z})^2 \tag{39}$$

where Θ represents the network parameters, Z represents the ground truth, and n is the number of pixels. For $\mathcal{L}_f$, the denoised result and ground truth are first transformed into the frequency domain, including three high-frequency components and one low-frequency component. Then, the MSE formula is used to calculate the error between the corresponding frequency components, and the final frequency loss can be defined as

$$[\mathbf{X}_{hh}, \mathbf{X}_{lh}, \mathbf{X}_{hl}, \mathbf{X}_{ll}] = \text{DWT}(\mathbf{Y}, \Theta) \tag{40}$$

$$[\mathbf{Z}_{hh}, \mathbf{Z}_{lh}, \mathbf{Z}_{hl}, \mathbf{Z}_{ll}] = \text{DWT}(\mathbf{Z}) \tag{41}$$

$$\mathcal{L}_f = \sum_{i=1}^{n} (\mathbf{X}_{hh} - \mathbf{Z}_{hh})^2 + \sum_{i=1}^{n} (\mathbf{X}_{lh} - \mathbf{Z}_{lh})^2 + \sum_{i=1}^{n} (\mathbf{X}_{hl} - \mathbf{Z}_{hl})^2 + \sum_{i=1}^{n} (\mathbf{X}_{ll} - \mathbf{Z}_{ll})^2 \tag{42}$$

where DWT represents the discrete wavelet transform operation, and $\mathbf{X}_{hh}, \mathbf{X}_{lh}, \mathbf{X}_{hl}, \mathbf{X}_{ll}$, and $\mathbf{Z}_{hh}, \mathbf{Z}_{lh}, \mathbf{Z}_{hl}, \mathbf{Z}_{ll}$, respectively, represent the frequency components of the denoised result and ground truth.

IV. Experiments

A. Experimental Settings

1) Satellite Hyperspectral Dataset: Each hyperspectral channel has a limited sensitivity to surface reflectance energy and is susceptible to interference from weather-related factors. Therefore, noise pollution is an inevitable challenge. In our research, we have selected data from imaging spectrometers installed on three different satellites, covering the same area at Yellow River Estuary: the U.S. EO-1 satellite with its Hyperion sensor, the Chinese GF-5 satellite, and the ZY-1 02 D satellite, to validate the performance of SUC-MSDN.

1) *GF-5 Dataset:* GF-5 satellite was launched on May 9, 2018. It is equipped with a hyperspectral imaging spectrometer covering the electromagnetic spectrum from 390 to 2504 nm, providing a spatial resolution of 30 m. This instrument acquires 150 bands in the visible and near-infrared range and 180 bands in the short-wave infrared range. The spectral resolution is finely detailed at 5 nm. For our study, the data used were captured on November 1, 2018, and obtained from the China Resources Satellite Application Center (https://data.cresda.cn). The image mainly shows Gaussian noise, impulse noise, stripes, and deadlines.
2) *EO-1 Hyperion Dataset:* The EO-1 satellite, launched on November 21, 2000, is equipped with the Hyperion hyperspectral imaging spectrometer. This instrument captures continuous surface images in the spectral range of 356–2577 nm. Hyperion offers 70 bands in the visible and near-infrared range and 172 bands in the short-wave infrared range. The spectral resolution is 10 nm, and it shares the same spatial resolution as GF-5. For our study, we obtained a Level 1 R product captured on April 5, 2012, from the U.S. Geological Survey (USGS) website (http://landsat.usgs.gov). The image mainly shows Gaussian noise and impulse noise.
3) *ZY-1 02 D Dataset:* The ZY-1 02 D satellite, launched on September 12, 2019, is equipped with a hyperspectral

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imaging spectrometer capable of capturing data in the spectral range of 395–2501 nm. The instrument comprises 76 bands in the visible and near infrared range and 90 bands in the short-wave infrared range, offers a spatial resolution of 30 m and a spectral resolution of 10 nm. For our study, we obtained data captured on December 11, 2019, from the China Resources Satellite Application Center (https://data.cresda.cn). The image predominantly exhibits Gaussian noise, stripes, and deadlines.

2) UAV Datasets: Deep learning models rely heavily on the number of training samples; however, obtaining a substantial amount of clean-noisy satellite HSI samples presents a significant challenge. To meet this challenge and create synthetic datasets for model training, we collected a large amount of publicly available UAV data considering that UVA data are less affected by atmospheric interference, which includes the following datasets: Xiongan dataset, Pavia University (PU), Pavia Center (PC) datasets, Houston dataset, Indian Pine dataset, Chikusei dataset.

3) Implementation and Model Training: To train SUC-MSDN, the Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.99$ is used, with a learning rate set to 1e-4, and a batch size of 8. Additionally, SUC-MSDN is trained on a Windows 10 X64 computer with 64-GB CPU memory and 1 × GPU GeForce GTX 3060Ti. It is important to note that the training process for each case is crucial. Initially, Case-1 is trained for 200 epochs. Then, the model trained in Case-1 is used to train Case-2, with the number of epochs set to 50. Similarly, Case-3 is also trained on the model from Case-1, using 50 epochs. Case-4 involves training SUC-MSDN for 200 epochs. Finally, Case-5 continues training on the model from Case-4 for an additional 50 epochs. It is worth noting that we have collected 1364 clean satellite HSI patches, each with 128 × 128 × 25 pixels. These clear patches are used as a ground truth, and the simulated noisy images are obtained by adding noise to these clear patches. These image pairs are then input into the pretrained model for network parameter fine-tuning. The fine-tuned model is subsequently used for denoising tests on satellite HSI.

4) Compared Methods: To evaluate the proposed SUC-MSDN, we compared it with a variety of state-of-the-art hyperspectral denoising methods, including filter-based methods such as BM3D [22], BM4D [23]; matrix factorization-based methods such as PARAFAC [11], hyper-Laplacian regularized unidirectional LR Tensor recovery method (LLRT) [47], FastHyDe [48]; and deep learning-based methods such as spectral enhanced rectangle transformer (SERT) [37], deep spatial–spectral global reasoning network (DSSGRN) [49], deep spatial–spectral representation learning (DSSRL) [50], low-rank network (LRNET) [51]). All parameters for these algorithms are derived from the optimal settings in their respective original references.

5) Quantitative Evaluation Metrics: To quantify the experimental results, several evaluation metrics are used [52], including peak signal-to-noise ratio (PSNR), structural similarity (SSIM), root mean square error (RMSE), spectral angle map (SAM), and relative global synthesis error (ERGAS). Higher PSNR and SSIM values typically indicate more satisfactory denoising results, with an ideal SSIM value of 1. Smaller RMSE, SAM, and ERGAS values represent better denoising performance, with ideal values of 0 for these metrics.

B. Experimental Results

1) Experimental Results of Simulated Datasets: Fig. 7 shows the denoising performance of various methods in different case. In Case-1, BM3D, LLRT, and LRNET are unable to effectively remove all Gaussian noise, and the denoised results of FastHyDe and PARAFAC have some color differences compared to the reference image. Other methods achieve good visualization results. SAM maps show different spectral distortion in these methods, BM3D and LLRT obtain severe spectral distortion, and our method exhibits comparative performance compared to all benchmark methods. In Case-2, all methods are proficient in eliminating Gaussian and impulse noise. However, there are differences between spectral and spatial fidelity. BM3D, LLRT, and PARAFAC demonstrate pronounced spectral distortion, while BM4D smooths spatial details while effectively reducing noise. Among the other methods, SUC-MSDN achieves superior denoising performance, producing the most favorable SAM map results, which represent minimal spectral reconstruction errors and well-preserved spatial details. In Case-3, LLRT yields the least satisfactory visual results, characterized by severe spectral distortion and an inability to eliminate stripes. BM3D, BM4D, and PARAFAC, while showing better visual results, still have significant stripes spectral distortion in SAM maps, indicating their limited effectiveness in stripes removal. Compared to DSSGRN, DSSRL, and LRNET, SERT and FastHyDe have some degree of spectral distortion, highlighting the superior performance of deep learning methods over traditional ones, with our method achieving the highest denoising performance among deep learning methods. In Case-4, all methods effectively eliminate Gaussian, impulse, and deadlines, but they show varying levels of reconstruction errors. Notably, BM3D exhibits a severe loss of spatial information and produces the poorest visual results. BM4D and FastHyDe tend to excessively smooth the denoised images. SERT still shows a small amount of residual Gaussian noise. Among these methods, PARAFAC, DSSGRN, DSSRL, LRNET, and SUC-MSDN achieve the most favorable visual results. However, SAM maps show that SUC-MSDN exhibits the least spectral distortion. In Case-5, although BM3D and BM4D successfully reduce most of the noise, they achieve the least satisfactory visual results, characterized by severe spectral and spatial distortion. LLRT, SERT, FastHyDe, and PARAFAC struggle to effectively eliminate certain instances of stripes and deadlines. In this case, all deep learning methods outperform traditional methods in performance, with SUC-MSDN exhibiting the least spectral reconstruction error.

To ensure an equitable comparison of the denoising performance of various methods across different cases, Table I presents the quantitative evaluation results. BM3D, BM4D, and LLRT consistently achieve lower quantitative evaluation scores across all cases. SERT and FastHyDe, on the other hand, show relatively superior performance when compared to other traditional methods, maintaining consistent and stable

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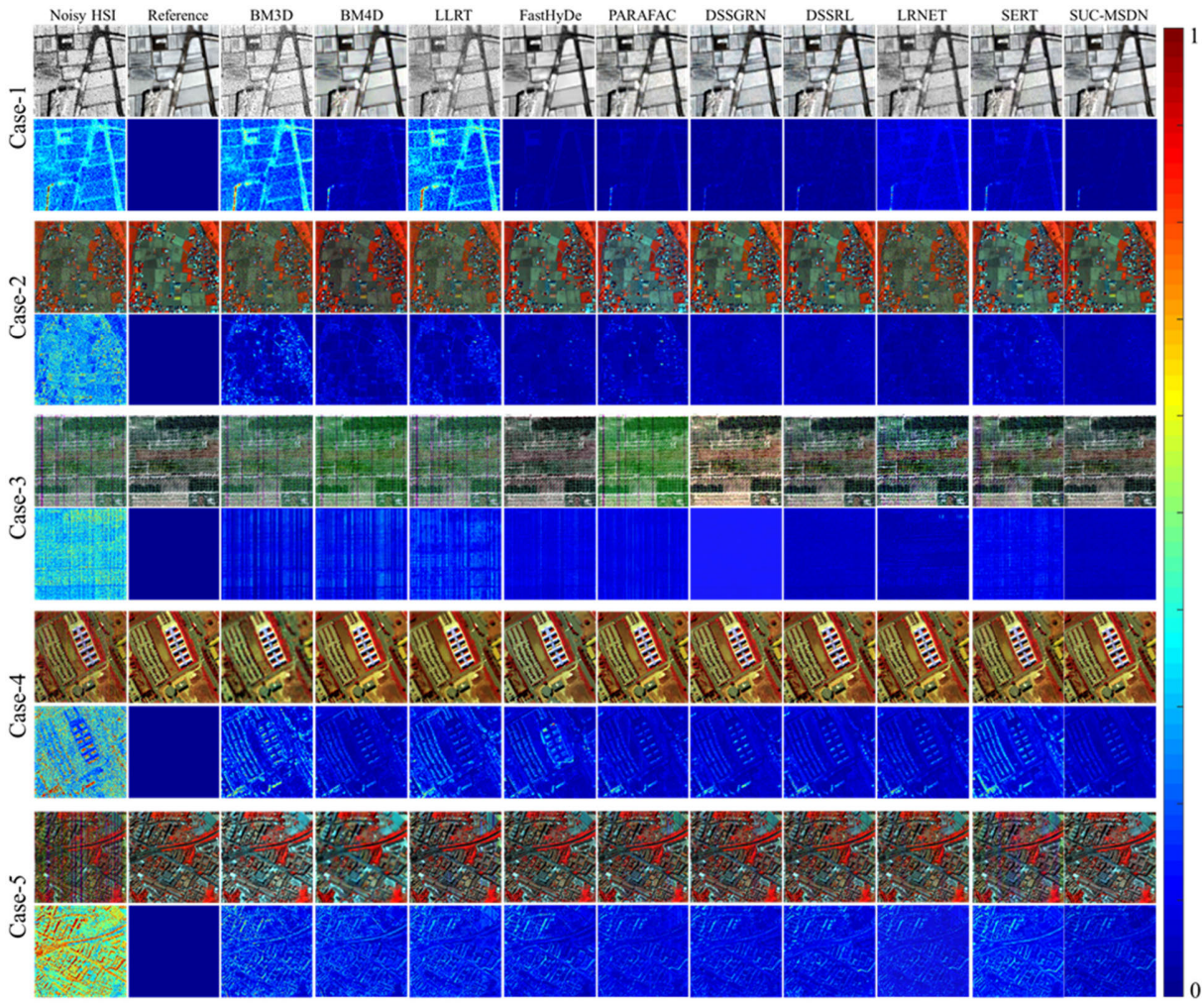


Fig. 7. Denoised results of different methods on different datasets.

results. Notably, all deep learning methods outperform traditional methods, showing better results across various cases, with SUC-MSDN consistently delivering the best results across all evaluation metrics.

Furthermore, Fig. 8 illustrates PSNR, SSIM, and cross correlation (CC) values of various methods in each spectral band for different cases. BM3D, BM4D, and LLRT consistently obtain the lowest values across almost all spectral bands in all cases. In contrast, our method demonstrates remarkable robustness across different cases, consistently achieving near-optimal values in almost all spectral bands. Fig. 9 presents spectral reflectance difference vectors for different land cover types, where SUC-MSDN shows the closest similarity to real land cover types.

2) Experimental Results of Real Datasets: To comprehensively evaluate SUC-MSDN model, we conduct denoising experiments on real hyperspectral datasets by fine-tuning the model.

1) *GF-5 Dataset:* In our experiments, we use a GF-5 image with dimensions of 232 × 992 × 275 to evaluate denoising performance. Fig. 10(a) shows the denoising results of various methods applied to the GF-5 hyperspectral dataset. The false-color image is constructed from bands 275, 274, and 1, which are heavily contaminated by Gaussian noise, impulse noise, stripes, and deadlines. BM3D and BM4D successfully remove a substantial portion of the noise; however, they tend to over-smooth the spatial information. LLRT, FastHyDe, and PARAFAC show significant spectral distortions, while SERT struggles to eliminate specific instances of stripes and deadlines. DSSGRN, DSSRL, and LRNET show limited effectiveness in addressing stripes and deadlines when compared to SUC-MSDN, and obtain some spectral and spatial distortions. Fig. 10(b) shows the denoising results for band 275, which is the band most affected by mixed noise. Deep learning methods continue to have a significant advantage over traditional methods, with SUC-MSDN delivering the most favorable visual results compared to others.
2) *EO-1 Hyperion Dataset:* The EO-1 Hyperion dataset used in our experiment has dimensions of 232 × 992 × 100. The denoising results of various methods are shown in Fig. 11(a). The false-color image is created from bands 100, 99, and 1, which are heavily contaminated by Gaussian and impulse noise. LLRT; unfortunately, only managed to eliminate a small portion of the noise,

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TABLE I
QUANTITATIVE RESULTS

Methods	BM3D	BM4D	LLRT	FastHyDe	PARAFAC	DSSGRN	DSSRL	LRNET	SERT	SUC-MSDN
Case 1: Gaussian noise										
PSNR	22.77	29.08	21.12	32.33	25.20	33.55	31.78	26.66	30.70	**37.37**
SAM	15.19	7.94	17.07	3.10	11.12	2.79	3.65	9.64	6.24	**1.29**
ERGAS	36.06	15.76	31.31	8.77	25.59	14.04	11.17	16.44	8.50	**5.81**
RMSE	0.100	0.027	0.102	0.025	0.032	0.021	0.017	0.051	0.024	**0.015**
SSIM	0.813	0.941	0.873	0.969	0.830	0.982	0.981	0.891	0.974	**0.986**
CC	0.937	0.960	0.977	0.977	0.929	0.991	0.982	0.882	0.985	**0.997**
Case 2: Gaussian noise + impulse noise										
PSNR	26.28	33.45	31.93	36.96	33.12	35.94	35.75	36.16	34.47	**37.91**
SAM	4.50	2.33	3.20	1.96	3.34	2.66	2.71	2.42	3.12	**1.84**
ERGAS	28.18	10.14	13.69	7.75	13.39	9.05	8.77	11.64	11.26	**6.85**
RMSE	0.047	0.024	0.025	0.014	0.021	0.016	0.017	0.016	0.018	**0.013**
SSIM	0.731	0.934	0.899	0.979	0.894	0.966	0.963	0.968	0.919	**0.988**
CC	0.821	0.978	0.963	0.996	0.964	0.992	0.994	0.978	0.974	**0.997**
Case 3: Gaussian noise + impulse noise + deadlines										
PSNR	27.19	29.64	29.10	29.29	27.57	31.89	31.07	31.14	30.59	**33.95**
SAM	6.04	5.49	6.71	5.04	5.23	3.18	3.36	2.85	6.81	**2.95**
ERGAS	12.94	8.31	10.56	10.15	11.50	7.32	7.88	7.34	11.18	**6.10**
RMSE	0.039	0.027	0.037	0.023	0.027	0.017	0.019	0.017	0.034	**0.014**
SSIM	0.870	0.933	0.916	0.934	0.927	0.963	0.959	0.961	0.896	**0.972**
CC	0.902	0.957	0.911	0.966	0.959	0.983	0.982	0.981	0.954	**0.989**
Case 4: Gaussian noise + impulse noise + stripes										
PSNR	25.23	27.45	28.66	37.04	27.84	37.07	36.78	33.58	33.44	**39.15**
SAM	14.73	12.10	7.64	3.98	10.68	4.23	4.24	5.13	6.67	**3.16**
ERGAS	36.83	31.57	21.44	8.72	24.18	8.98	9.16	14.24	13.27	**7.34**
RMSE	0.056	0.081	0.057	0.014	0.043	0.014	0.015	0.022	0.022	**0.010**
SSIM	0.745	0.929	0.821	0.958	0.729	0.962	0.961	0.937	0.905	**0.967**
CC	0.855	0.961	0.952	0.990	0.915	0.990	0.992	0.974	0.974	**0.995**
Case 5: Gaussian noise + impulse noise + deadlines + stripes										
PSNR	22.76	27.17	29.69	29.69	23.10	32.39	31.66	33.91	31.30	**34.12**
SAM	8.66	5.40	7.57	7.57	7.35	5.39	6.29	5.66	8.31	**3.28**
ERGAS	38.37	15.30	26.21	26.21	36.49	17.62	17.26	16.33	21.82	**10.62**
RMSE	0.082	0.020	0.033	0.033	0.081	0.026	0.026	0.020	0.027	**0.017**
SSIM	0.725	0.966	0.885	0.885	0.758	0.958	0.948	0.963	0.927	**0.976**
CC	0.834	0.983	0.937	0.937	0.864	0.985	0.980	0.977	0.951	**0.991**

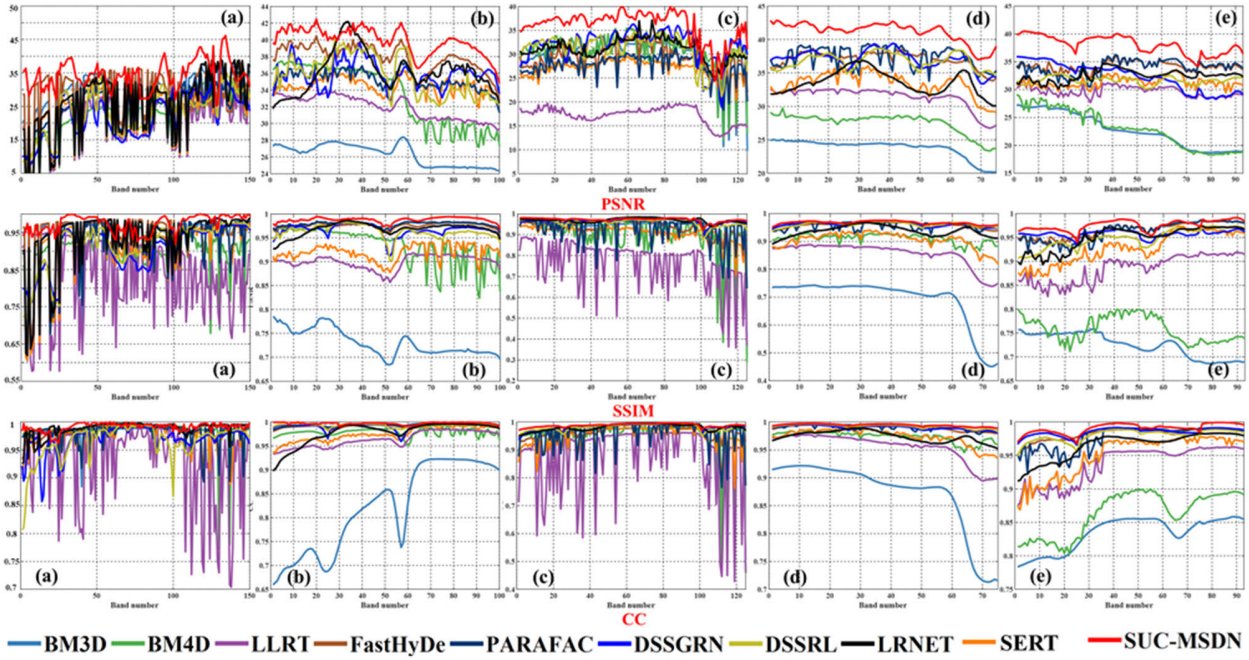


Fig. 8. Quantitative evaluation results of PSNR, SSIM, and CC for each band in different cases. (a) Case-1, (b) Case-2, (c) Case-3, (d) Case-4, and (e) Case-5.

result in the least satisfactory visual results. BM3D and BM4D still have spatial smoothing tendencies, while other methods show more favorable visual results. Fig. 11(b) shows the denoising results of different

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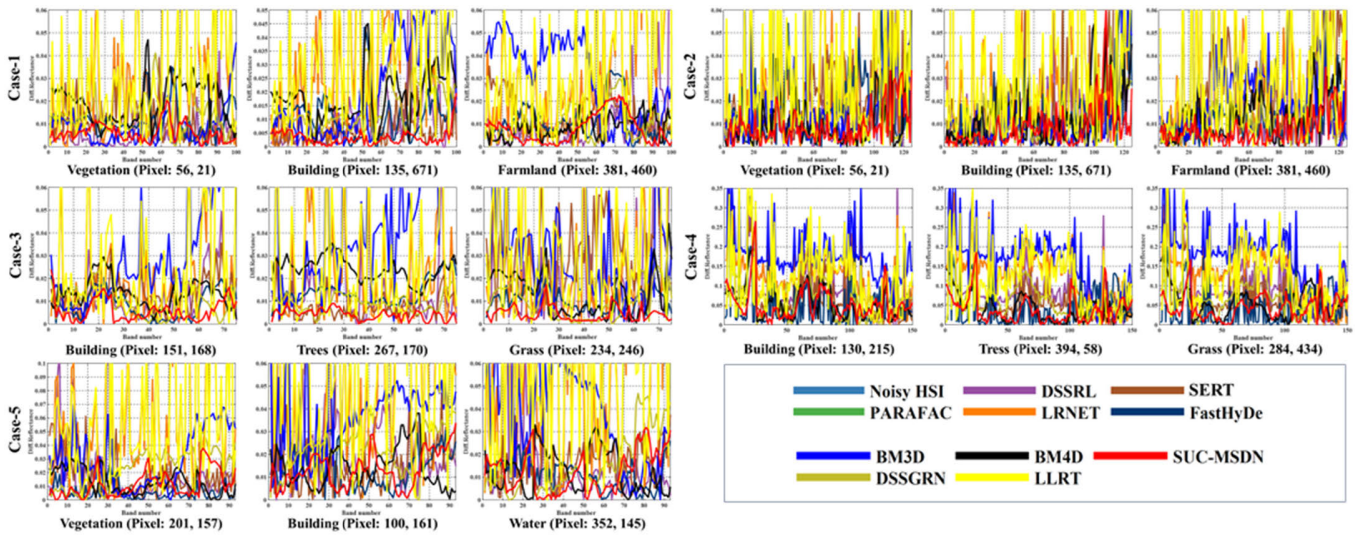


Fig. 9. Spectral reflectance difference vectors.

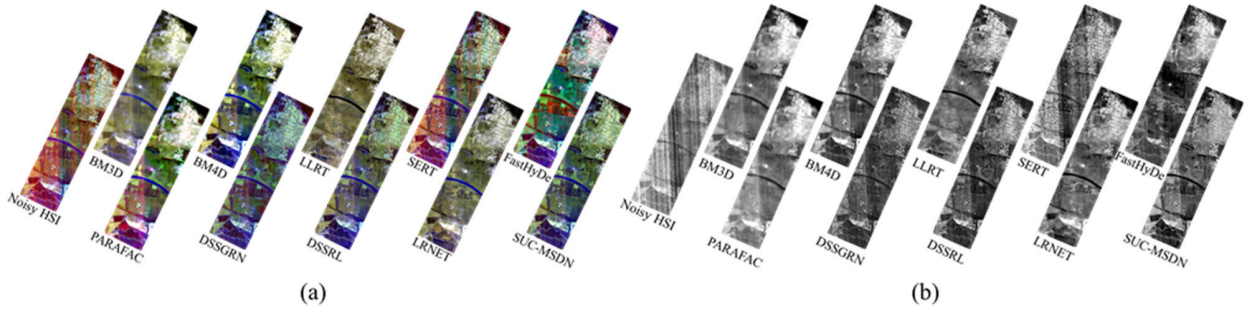


Fig. 10. Denoising results for the GF-5 dataset. (a) R: band 275, G: band 274, and B: band 1. (b) Band 275.

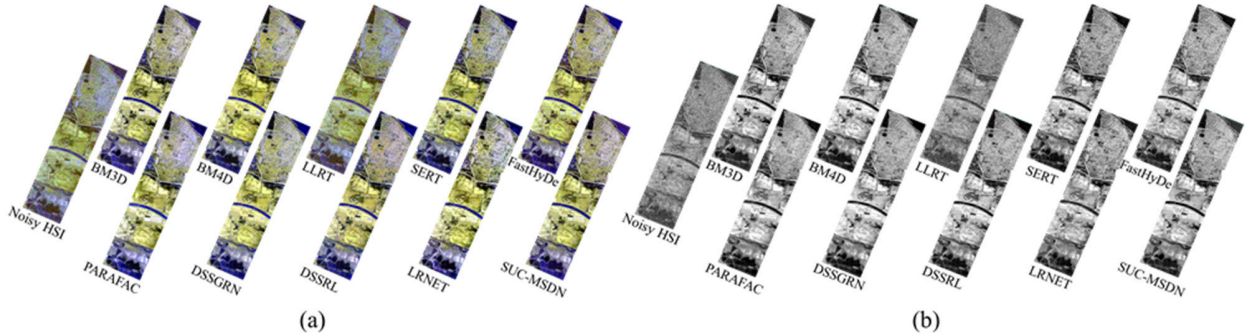


Fig. 11. Denoising results for the EO-1 Hyperion dataset. (a) R: band 100, G: band 99, and B: band 1. (b) Band 100.

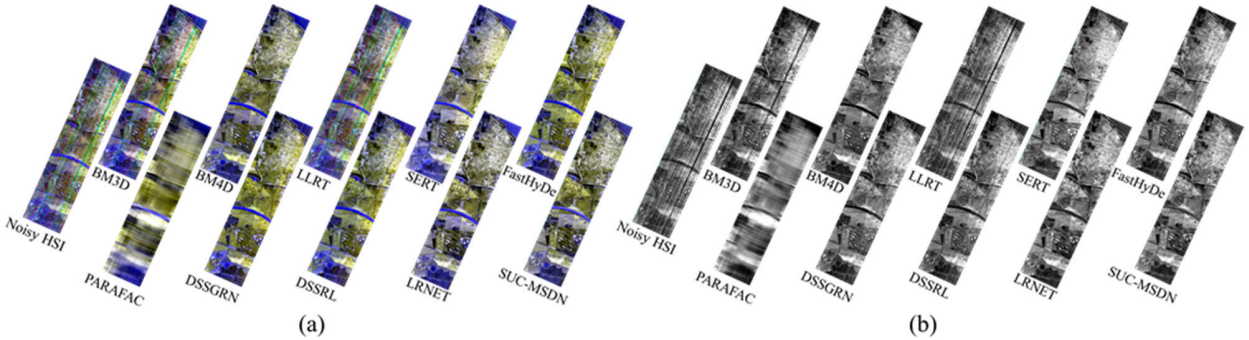


Fig. 12. Denoising results for the ZY-1 02 D dataset. (a) R: band 125, G: band 124, and B: band 1. (b) Band 125.

methods specifically for band 100, which is the band most affected by noise. With the exception of LLRT, almost all methods demonstrate significant denoising capabilities.

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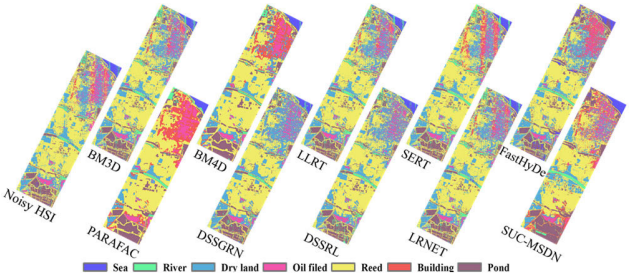


Fig. 13. Classification results using SVM before and after the denoising of GF-5 dataset.

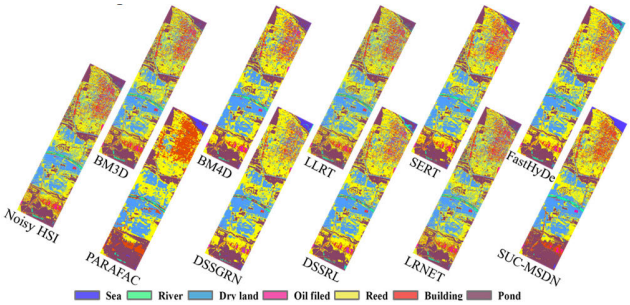


Fig. 14. Classification results using SVM before and after the denoising of EO-1 Hyperion dataset.

3) *ZY-1 02 D Dataset:* The ZY-1 02 D hyperspectral dataset consists of 232 × 992 × 125 pixels. In Fig. 12(a), a false-color denoised image, synthesized from bands 125, 124, and 1, is presented. These bands are heavily affected by Gaussian noise, strip noise, and deadlines. BM3D and LLRT struggle to effectively eliminate strip and deadlines. PARAFAC causes a significant loss of spatial information, and BM4D and SERT show some spectral distortions. However, other methods obtain satisfactory visual results. Fig. 12(b) shows the denoising results of different methods, specifically for band 125, which is the band most affected by noise. With the exception of BM3D, LLRT, and PARAFAC, which have poor results, BM4D has lost spatial information in some regions. Also, FastHyDe and DSSRL still cannot completely remove some strip and deadlines. On the other hand, DSSGRN, LRNET, and SUC-MSDN produce the most favorable results.

3) Land-Cover Mapping: To assess the real-world applicability of denoising results, we conduct land cover classification experiments, including both the original noisy images and denoised images obtained through various methods. Support vector machine (SVM) classifier is used in these experiments to generate land cover maps. Ground truth samples, covering categories, such as sea, river, dry land, oil field, reed, building, and pond, are selected using Google Earth images and prior research conducted by Sun et al. [53]. In the classification process, 20% of the samples are allocated for training the classifier, while the remaining samples are used for accuracy testing. Specifically, we calculate the overall classification accuracy (OA) and Kappa coefficient (KC) to assess the accuracy of classification maps.

Figs. 13–15 illustrate the classification results for three datasets, and Table II provides an overview of the classification

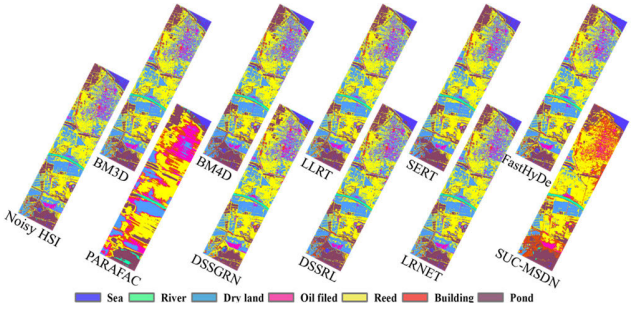


Fig. 15. Classification results using SVM before and after the denoising of ZY-1 02 D dataset.

accuracy. The experimental results show that the classification results of the original images are significantly affected by noise. Denoised images results in a noticeable improvement in classification accuracy when compared to original images, highlighting the importance of image denoising in practical applications. BM3D, LLRT, and PARAFAC achieve limited denoising performance, resulting in only marginal improvements in accuracy for land cover classification when compared to the original images. In contrast, other methods effectively improve HSI classification accuracy, with deep learning methods showing a notable advantage. In conclusion, due to the superior denoising performance of SUC-MSDN, it consistently delivers the best classification results across experiments.

C. Discussion

1) Multistage Ablation Experiments: In this article, we propose a multistage denoising network, SUC-MSDN, which leverages low-scale denoised images as prior information for high-scale denoising, resulting in satisfactory denoising performance. Therefore, the rational setting of stages is an important factor affecting HSI denoising. In the experiments, we use 128 × 128 patches to train the network and analyze the denoising performance of different stage settings. We set the number of stages to [1, 2, 3, 4] and keep all network parameters the same. Fig. 16 and Table III show the image denoising results with different stage settings in Case-5. When only one stage is set, denoised images have severe spectral and spatial distortions. As the number of stages increases, denoising performance gradually improves, with the best result achieved in the third stage. Experimental results show that a multistage setting can effectively enhance denoising performance.

2) Network Module Ablation Experiment: SUC-MSDN explores the spatial–spectral distribution characteristics and texture distribution characteristics of noise, achieving precise noise separation through feature interaction. To validate the contributions of the spatial–spectral feature extraction branch and the texture feature extraction branch to SUC-MSDN, ablation experiments were carried out. Fig. 17 and Table IV illustrate denoising performance when using only the spatial–spectral feature extraction network, only the texture feature extraction network, and their combined application. Experimental results show that both the spatial–spectral

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TABLE II
CLASSIFICATION ACCURACY

Methods	Noisy HSI	BM3D	BM4D	LLRT	FastHyDe	PARAFAC	DSSGRN	DSSRL	LRNET	SERT	SUC-MSDN
GF-5 dataset											
OA (%)	79.94	81.61	89.22	83.65	85.63	71.89	89.71	90.66	81.80	85.69	94.50
Kappa	0.744	0.766	0.861	0.777	0.816	0.677	0.881	0.893	0.769	0.818	0.923
EO-1 Hyperion dataset											
OA (%)	75.95	75.78	78.97	74.06	76.78	67.51	80.21	78.38	75.92	76.66	87.32
Kappa	0.652	0.649	0.695	0.625	0.670	0.581	0.718	0.697	0.652	0.668	0.819
ZY-1 02 D dataset											
OA (%)	78.54	77.26	79.68	78.98	83.64	61.34	85.56	84.66	78.87	81.59	89.77
Kappa	0.687	0.681	0.705	0.690	0.747	0.516	0.779	0.761	0.689	0.731	0.816

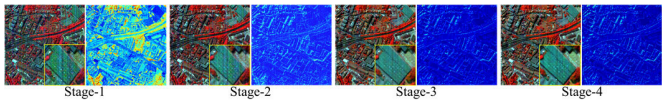


Fig. 16. Multistage ablation experiments.

TABLE III
MULTISTAGE ABLATION EXPERIMENTS

Methods	PSNR	SAM
Stage-1	14.13	18.69
Stage-2	27.20	10.94
Stage-3	34.12	3.28
Stage-4	33.91	4.07

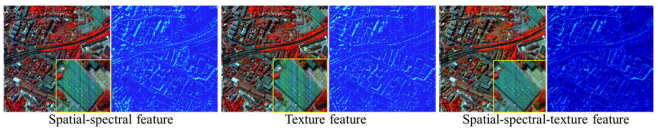


Fig. 17. Spatial–spectral-texture ablation experiment.

TABLE IV
SPATIAL-SPECTRAL-TEXTURE ABLATION EXPERIMENT

Methods	PSNR	SAM
Spatial-spectral features	28.09	9.88
Texture features	28.97	9.04
Spatial-spectral-texture features	34.12	3.28

feature extraction network and the texture feature extraction network yield similar denoising results. However, a significant improvement in the denoising performance of SUC-MSDN is observed when spatial–spectral-texture information is simultaneously integrated. This improvement stems from the varying sensitivity of different types of noise signals to spatial–spectral and texture responses. Guided by the interaction between spatial–spectral and texture features, noise can be accurately extracted.

During the noise extraction process, due to the severe entanglement between noise signals and clear signals, the extracted noise signals typically contain clear signals. If these are directly removed, it can lead to a loss of spatial details in the denoised images. In our experiments, we designed a NAM to optimize noise features. To verify the rationality of the NAM design, we conducted ablation studies on NAM. Fig. 18 and

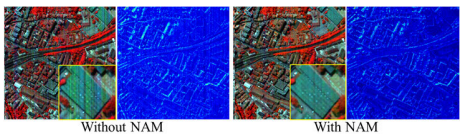


Fig. 18. NAM ablation experiment.

TABLE V
NAM ABLATION EXPERIMENTS

Methods	PSNR	SAM
Without NAM	29.68	8.11
With NAM	34.12	3.28

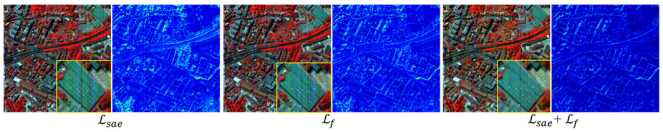


Fig. 19. Loss function ablation experiment.

TABLE VI
LOSS FUNCTION ABLATION EXPERIMENT

Methods	PSNR	SAM
$\mathcal{L}_{sae}$	24.37	11.20
$\mathcal{L}_f$	26.58	10.81
$\mathcal{L}_{sae}+\mathcal{L}_f$	34.12	3.28

Table V demonstrate that NAM can effectively enhance the denoising performance of SUC-MSDN.

3) Loss Function Ablation Experiment: To achieve better denoising performance, SUC-MSDN uses a combination of $\mathcal{L}_{\text{sae}}$ and $\mathcal{L}_f$ loss functions during network training. This combination constrains the network from both spatial–spectral and frequency perspectives, further improving the ability of network to obtain optimal results. In the experiments, we verify the impact of different loss functions on network performance. Fig. 19 and Table VI show the denoising results when using only $\mathcal{L}_{\text{sae}}$, only $\mathcal{L}_f$, and the combination of both $(\mathcal{L}_{\text{sae}} + \mathcal{L}_f)$. When using only $\mathcal{L}_{\text{sae}}$ or $\mathcal{L}_f$, the PSNR decreases significantly, and spectral errors increase compared to using $\mathcal{L}_{\text{sae}}+ \mathcal{L}_f$. This demonstrates that combining $\mathcal{L}_{\text{sae}}$ and $\mathcal{L}_f$ effectively enhances network-training performance.

4) Sample Generation Strategy Ablation Experiment: Currently, a significant portion of HSI denoising research relies on training networks with synthetic data. However, there is a

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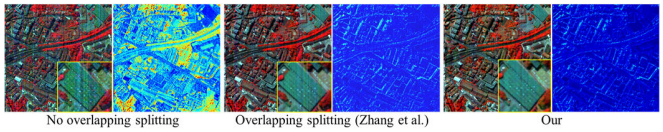


Fig. 20. Sample generation strategy ablation experiment.

TABLE VII
SAMPLE GENERATION STRATEGY ABLATION EXPERIMENT

Methods	PSNR	SAM
No overlapping splitting	11.06	25.36
Overlapping splitting [51]	30.62	7.04
Our	34.12	3.28

lack of standardized methods for generating these synthetic samples. Recognizing that sample quality can significantly affect the rapid convergence of networks and overall denoising performance, we propose a novel sample generation strategy. This strategy aims to ensure accurate denoising while avoiding network overfitting.

In summary, previous sample generation strategies can be broadly categorized into two groups: the "No Overlap Splitting Strategy" (where samples are created without overlap in terms of spatial and spectral ranges during data slicing) and the "Overlap Splitting Strategy" (which involves defining spatial and spectral strides, allowing for overlapping samples during data slicing). To validate the effectiveness of our proposed sample generation strategy, we compare these two existing strategies.

Fig. 20 and Table VII present the results of models trained with samples generated using different strategies. The experimental results clearly indicate that our strategy outperforms current sample generation strategies. This superiority may be attributed to inconsistencies in spectral ranges between training and test samples introduced by the other two strategies, potentially leading to reduced network generalization.

5) Computational Efficiency: SUC-MSDN uses denoised images from the previous stage as prior information for the next stage, leading to more precise noise removal. This method not only enhances network interpretability but also mitigates the computational burden associated with feature stacking. To evaluate the computational efficiency of different methods, we measure computation times for various methods using real datasets. For deep learning methods, the training times correspond to the training periods for Case-1, Case-4, and Case-5.

Table VIII provides an overview of training and testing times for different methods. In denoising tests, deep learning methods outperform traditional methods in terms of computational efficiency, with SUC-MSDN achieving the best computational efficiency. Regarding training times, DSSGRN and DSSLR, which use the UNet framework for network training, significantly reduce network-training speeds. In contrast, SUC-MSDN outperforms LRNET and SERT and achieves the highest computational efficiency.

6) Analysis of Network Convergence Rate: Fig. 21 shows the SSIM values at different stages of SUC-MSDN across all cases. For Case-1, Stage-1 peaked around the 50th epoch, Stage-2 around the 80th epoch, and Stage-3 around the 110th

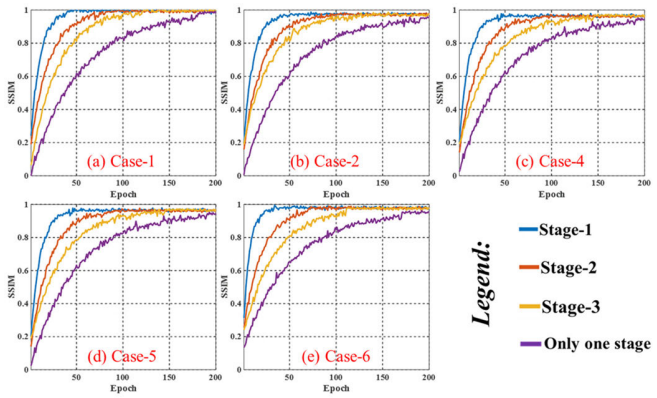


Fig. 21. Convergence rates of SUC-MSDN at different stages in various cases.

TABLE VIII
RUN TIMES FOR THE DIFFERENT HSI DENOISING METHODS CONDUCTED ON THESE REAL DATASETS

Methods	GF-5 dataset		EO-1 Hyperion dataset		ZY-1 02 D dataset	
	Train (min) *Case 5*	Test (min)	Train (min) *Case 1*	Test (min)	Train (min) *Case 4*	Test (min)
BM3D	—	20.64	—	8.26	—	12.67
BM4D	—	29.71	—	15.39	—	20.05
LLRT	—	56.89	—	27.34	—	37.64
FastHyDe	—	17.26	—	7.24	—	11.08
PARAFAC	—	42.24	—	22.65	—	27.61
DSSGRN	3067	0.71	3108	0.46	3044	0.58
DSSRL	3221	0.77	3297	0.61	3158	0.60
LRNET	2974	0.67	3067	0.49	3073	0.51
SERT	4657	0.85	4903	0.91	4257	0.94
SUC-MSDN	2321	0.25	2259	0.14	2396	0.20

epoch. Cases 2 and 3 show similar patterns, with Stage-1 peaking around the 60th epoch, Stage-2 around the 110th epoch, and Stage-3 around the 130th epoch. In Case 4, Stage-1 peaks around the 75th epoch, Stage-2 peaks around the 100th epoch, and Stage-3 at around the 120th epoch. For Case 5, Stage-1 peaks around the 90th epoch, Stage-2 around the 120th epoch, and Stage-3 around the 150th epoch. It is worth noting that Stage-1 has the fastest convergence rate among all cases, while Stage-3 is the slowest. This effectively demonstrates that noise in low-scale images is more easily removed, providing a valid argument for the rationale behind the multistage setup. Additionally, in all cases, only one training stage results in a significantly slow convergence rate, and it cannot reach the optimal value within a limited number of epochs. This indicates that the multistage setting effectively enhances the convergence rate of the network and quickly achieves optimal results.

V. CONCLUSION

In the article, we propose a novel multistage fusion-denoising network for satellite HSI. Existing deep learning methods often overlook the inherent relationships between multiscale features. Simple feature stacking can result in a heavy computational burden. To solve this problem and fully consider the distribution characteristics of

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noise signals at different scales, we propose SUC-MSDN. SUC-MSDN explored the spatial–spectral-texture distribution characteristics of noise, and a novel NAM was constructed for optimizing noise features. It incorporates noise separation modules between different scales, effectively using small-scale denoised images as prior information to enhance denoising performance of large-scale noisy images. This method not only mitigates the high-dimensional computational burden caused by feature stacking but also enhances the interpretability of network. Extensive simulated and real experiments show that SUC-MSDN can achieve satisfactory results in HSI denoising. However, how to transfer a network trained on UAV data without the involvement of satellite hyperspectral samples remains a challenge that SUC-MSDN needs to address. In the future, we will explore the integration of network transfer mechanisms with SUC-MSDN to further enhance the applicability of network.

REFERENCES

[1] W. Sun, K. Ren, X. Meng, C. Xiao, G. Yang, and J. Peng, "A band divide-and-conquer multispectral and hyperspectral image fusion method," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5502113.

[2] R. Jafari and M. M. Lewis, "Arid land characterisation with EO-1 hyperion hyperspectral data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 19, pp. 298–307, Oct. 2012.

[3] W. He, Q. Yao, C. Li, N. Yokoya, and Q. Zhao, "Non-local meets global: An integrated paradigm for hyperspectral denoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6861–6870.

[4] J. Li, K. Zheng, W. Liu, Z. Li, H. Yu, and L. Ni, "Model-guided coarse-to-fine fusion network for unsupervised hyperspectral image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[5] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.

[6] Q. Yuan, L. Zhang, and H. Shen, "Hyperspectral image denoising employing a spectral–spatial adaptive total variation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3660–3677, Oct. 2012.

[7] W. Wei, L. Zhang, Y. Jiao, C. Tian, C. Wang, and Y. Zhang, "Intracluster structured low-rank matrix analysis method for hyperspectral denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 866–880, Feb. 2019.

[8] Y. Chen, X. Cao, Q. Zhao, D. Meng, and Z. Xu, "Denoising hyperspectral image with non-i.I.d. noise structure," *IEEE Trans. Cybern.*, vol. 48, no. 3, pp. 1054–1066, Mar. 2018.

[9] Q. Zhang, Y. Zheng, Q. Yuan, M. Song, H. Yu, and Y. Xiao, "Hyperspectral image denoising: From model-driven, data-driven, to Model-Data-driven," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 10, pp. 13143–13163, 2023.

[10] Q. Zhang, Y. Dong, Q. Yuan, M. Song, and H. Yu, "Combined deep priors with low-rank tensor factorization for hyperspectral image restoration," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[11] X. Liu, S. Bourennane, and C. Fossati, "Denoising of hyperspectral images using the PARAFAC model and statistical performance analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3717–3724, Oct. 2012.

[12] T. Pang, H. Zheng, Y. Quan, and H. Ji, "Recorrupted-to-recorrupted: Unsupervised deep learning for image denoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2043–2052.

[13] Q. Zhang, Q. Yuan, M. Song, H. Yu, and L. Zhang, "Cooperated spectral low-rankness prior and deep spatial prior for HSI unsupervised denoising," *IEEE Trans. Image Process.*, vol. 31, pp. 6356–6368, 2022.

[14] M. Mafi, H. Martin, M. Cabrerizo, J. Andrian, A. Barreto, and M. Adjouadi, "A comprehensive survey on impulse and Gaussian denoising filters for digital images," *Signal Process.*, vol. 157, pp. 236–260, Apr. 2019.

[15] H. Hwang and R. A. Haddad, "Adaptive median filters: New algorithms and results," *IEEE Trans. Image Process.*, vol. 4, no. 4, pp. 499–502, Apr. 1995.

[16] R. Hardie, "A fast image super-resolution algorithm using an adaptive Wiener filter," *IEEE Trans. Image Process.*, vol. 16, no. 12, pp. 2953–2964, Dec. 2007.

[17] I. W. Selesnick, H. L. Graber, D. S. Pfeil, and R. L. Barbour, "Simultaneous low-pass filtering and total variation denoising," *IEEE Trans. Signal Process.*, vol. 62, no. 5, pp. 1109–1124, Mar. 2014.

[18] F. M. Bayer, A. J. Kozakevicius, and R. J. Cintra, "An iterative wavelet threshold for signal denoising," *Signal Process.*, vol. 162, pp. 10–20, Sep. 2019.

[19] P. Jain and V. Tyagi, "LAPB: Locally adaptive patch-based wavelet domain edge-preserving image denoising," *Inf. Sci.*, vol. 294, pp. 164–181, Feb. 2015.

[20] S. H. Ismael, F. M. Mustafa, and I. T. Okümüs, "A new approach of image denoising based on discrete wavelet transform," in *Proc. World Symp. Comput. Appl. Res. (WSCAR)*, 2016, pp. 36–40.

[21] J. Bai and X.-C. Feng, "Image denoising using generalized anisotropic diffusion," *J. Math. Imag. Vis.*, vol. 60, no. 7, pp. 994–1007, Sep. 2018.

[22] S. Sarjanoja, J. Boutellier, and J. Hannuksela, "BM3D image denoising using heterogeneous computing platforms," in *Proc. Conf. Design Archit. Signal Image Process. (DASIP)*, Sep. 2015, pp. 1–8.

[23] M. Maggioni, V. Katkovnik, K. Egiazarian, and A. Foi, "Nonlocal transform-domain filter for volumetric data denoising and reconstruction," *IEEE Trans. Image Process.*, vol. 22, no. 1, pp. 119–133, Jan. 2013.

[24] H. Wang, Y. Li, Y. Cen, and Z. He, "Multi-matrices low-rank decomposition with structural smoothness for image denoising," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 2, pp. 349–361, Feb. 2020.

[25] M. Elad and M. Aharon, "Image denoising via sparse and redundant representations over learned dictionaries," *IEEE Trans. Image Process.*, vol. 15, no. 12, pp. 3736–3745, Dec. 2006.

[26] S. Meng, L.-T. Huang, and W.-Q. Wang, "Tensor decomposition and PCA jointed algorithm for hyperspectral image denoising," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 7, pp. 897–901, Jul. 2016.

[27] W. He, H. Zhang, L. Zhang, and H. Shen, "Hyperspectral image denoising via noise-adjusted iterative low-rank matrix approximation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3050–3061, Jun. 2015.

[28] M. Wang, J. Yu, J.-H. Xue, and W. Sun, "Denoising of hyperspectral images using group low-rank representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 9, pp. 4420–4427, Sep. 2016.

[29] X. Zeng, W. Bian, W. Liu, J. Shen, and D. Tao, "Dictionary pair learning on Grassmann manifolds for image denoising," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4556–4569, Nov. 2015.

[30] T. Lu, S. Li, L. Fang, Y. Ma, and J. A. Benediktsson, "Spectral–spatial adaptive sparse representation for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 373–385, Jan. 2016.

[31] J. Xue, Y. Zhao, W. Liao, and J. C. Chan, "Nonlocal low-rank regularized tensor decomposition for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 5174–5189, Jul. 2019.

[32] H. Zhang, L. Liu, W. He, and L. Zhang, "Hyperspectral image denoising with total variation regularization and nonlocal low-rank tensor decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3071–3084, May 2019.

[33] Y. Wu, L. Fang, and S. Li, "Weighted tensor Rank-1 decomposition for nonlocal image denoising," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 2719–2730, Jun. 2019.

[34] Q. Yuan, Q. Zhang, J. Li, H. Shen, and L. Zhang, "Hyperspectral image denoising employing a spatial–spectral deep residual convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1205–1218, Feb. 2019.

[35] A. Maffei, J. M. Haut, M. E. Paoletti, J. Plaza, L. Bruzzone, and A. Plaza, "A single model CNN for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2516–2529, Apr. 2019.

[36] J. Zhang, Z. Cai, F. Chen, and D. Zeng, "Hyperspectral image denoising via adversarial learning," *Remote Sens.*, vol. 14, no. 8, p. 1790, Apr. 2022.

[37] M. Li, J. Liu, Y. Fu, Y. Zhang, and D. Dou, "Spectral enhanced rectangle transformer for hyperspectral image denoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 5805–5814.

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[38] Y. Miao, L. Zhang, L. Zhang, and D. Tao, "DDS2M: Self-supervised denoising diffusion spatio-spectral model for hyperspectral image restoration," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 12086–12096.

[39] H. Sun, M. Liu, K. Zheng, D. Yang, J. Li, and L. Gao, "Hyperspectral image denoising via low-rank representation and CNN denoiser," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 716–728, 2022.

[40] Y.-C. Miao, X.-L. Zhao, X. Fu, J.-L. Wang, and Y.-B. Zheng, "Hyperspectral denoising using unsupervised disentangled spatiospectral deep priors," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5513916.

[41] F. Xiong, J. Zhou, Q. Zhao, J. Lu, and Y. Qian, "MAC-Net: Model-aided nonlocal neural network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519414.

[42] F. Xiong, J. Zhou, S. Tao, J. Lu, J. Zhou, and Y. Qian, "SMDS-Net: Model guided spectral-spatial network for hyperspectral image denoising," *IEEE Trans. Image Process.*, vol. 31, pp. 5469–5483, 2022.

[43] J. Li, K. Zheng, Z. Li, L. Gao, and X. Jia, "X-shaped interactive autoencoders with cross-modality mutual learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518317.

[44] C. Tian, L. Fei, W. Zheng, Y. Xu, W. Zuo, and C.-W. Lin, "Deep learning on image denoising: An overview," *Neural Netw.*, vol. 131, pp. 251–275, Nov. 2020.

[45] F. Jia, W. H. Wong, and T. Zeng, "DDUNet: Dense dense U-Net with applications in image denoising," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 354–364.

[46] J. Zhang, L. Sang, Z. Wan, Y. Wang, and Y. Li, "Deep convolutional neural network based on multi-scale feature extraction for image denoising," in *Proc. IEEE Int. Conf. Vis. Commun. Image Process. (VCIP)*, Dec. 2020, pp. 213–216.

[47] Y. Chang, L. Yan, and S. Zhong, "Hyper-Laplacian regularized unidirectional low-rank tensor recovery for multispectral image denoising," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5901–5909.

[48] L. Zhuang and J. M. Bioucas-Dias, "Fast hyperspectral image denoising and inpainting based on low-rank and sparse representations," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 730–742, Mar. 2018.

[49] X. Cao, X. Fu, C. Xu, and D. Meng, "Deep spatial–spectral global reasoning network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504714.

[50] W. Dong, H. Wang, F. Wu, G. Shi, and X. Li, "Deep spatial–spectral representation learning for hyperspectral image denoising," *IEEE Trans. Comput. Imag.*, vol. 5, no. 4, pp. 635–648, Dec. 2019.

[51] H. Zhang, H. Chen, G. Yang, and L. Zhang, "LR-net: low-rank spatial–spectral network for hyperspectral image denoising," *IEEE Trans. Image Process.*, vol. 30, pp. 8743–8758, 2021.

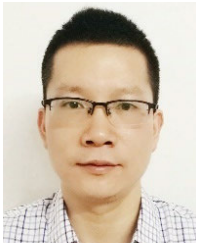
[52] Y. Chang, L. Yan, and W. Liao, "HSI-DeNet: Hyperspectral image restoration via convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 667–682, Feb. 2018.

[53] W. Sun et al., "A simple and effective spectral–spatial method for mapping large-scale coastal wetlands using China ZY1-02D satellite hyperspectral images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 104, Dec. 2021, Art. no. 102572.



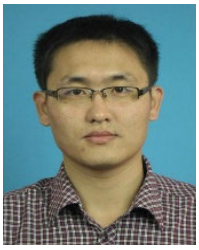
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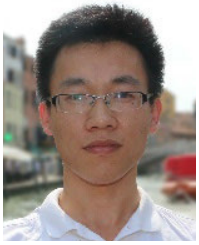
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